The reciprocal supernorm perimeter asymptotic

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Abstract

We prove Conjecture 3.6 of Lagarias and Sun by decomposing a partition according to its largest part. A positive generating function bounds the contribution from long residual partitions exponentially. Mertens' product theorem and the prime number theorem then give the required constant.

Task 1: the original problem

Write p_j for the jth prime, with $p_1 = 2$. For a nonempty partition $\lambda = (\lambda_1 \geq \cdots \geq \lambda_\ell \geq 1)$, put

$$\operatorname{per}(\lambda) = \lambda_1 + \ell - 1, \qquad \widehat{N}(\lambda) = \prod_{i=1}^{\ell} p_{\lambda_i}, \qquad \widehat{W}_{\mathrm{per}}(n) = \sum_{\operatorname{per}(\lambda)=n} \frac{1}{\widehat{N}(\lambda)}.$$

The source assigns the empty partition perimeter 1; its convention only changes a possible value at $n = 1$. All identities below concerning $\widehat{W}_{\mathrm{per}}$ are asserted for $n \geq 2$ and are unaffected.

Theorem 1. *As $n \to \infty$ through the positive integers,*

$$\widehat{W}_{\mathrm{per}}(n) \sim \frac{e^{\gamma}}{n},$$

where γ is Euler's constant.

This is the perimeter conjecture in both the 2023 preprint and the 2024 journal version of [1]. The latter already proves the separate maximum-part assertion as Theorem 3.5; we claim no new result for that assertion. We use the classical estimates

$$p_k \sim k \log k, \qquad \prod_{j=1}^{k} (1 - p_j^{-1})^{-1} \sim e^{\gamma} \log p_k. \tag{1}$$

These are consequences of Lemmas 4.3 and 4.4 of [1].

A coefficient decomposition and a uniform tail

For $k \geq 1$, define the length generating function

$$H_k(z) = \prod_{j=1}^{k} (1 - z/p_j)^{-1} = \sum_{r \geq 0} h_{k,r} z^r \qquad (|z| < 2). \tag{2}$$

Here $h_{k,r}$ is the sum of reciprocal supernorms over partitions with exactly r parts, all at most k. In particular $h_{k,r} \geq 0$ and $h_{k,r} \leq h_{k+1,r}$, coefficient by coefficient. The empty partition contributes $h_{k,0} = 1$.

Lemma 1. *For every $n \geq 2$,*

$$\widehat{W}_{\mathrm{per}}(n) = \sum_{r=0}^{n-1} \frac{h_{n-r,r}}{p_{n-r}}. \tag{3}$$

Proof. If the largest part is k, remove one copy of k. The remaining partition has $r = n - k$ parts, each at most k, and its supernorm is the original supernorm divided by p_k. Conversely, adjoining k to any such residual partition gives a unique partition of perimeter $k + r = n$. Repeated largest parts create no multiplicity: partitions are multisets and exactly one copy is removed or adjoined. Summing over $k = 1, \ldots, n$ proves the identity. □

For $n \geq 4$, set $m = \lfloor \sqrt{n} \rfloor$ and

$$T_n = \sum_{r>m} h_{n,r}, \qquad B_n = (n+1)^3 (2/3)^{m+1}.$$

We have

$$0 \leq T_n \leq B_n. \tag{4}$$

Indeed, positivity and evaluation of (2) at $z = 3/2$ give

$$T_n \leq (2/3)^{m+1} H_n(3/2).$$

Since $p_j \geq j + 1$, and, for every integer $j \geq 1$,

$$\frac{j+1}{j-1/2} \leq \left(\frac{j+1}{j}\right)^3 \quad \text{because} \quad (2j-1)(j+1)^2 - 2j^3 = 3j^2 - 1 > 0,$$

we obtain the telescoping bound

$$H_n(3/2) \leq \prod_{j=1}^{n} \frac{j+1}{j-1/2} \leq \prod_{j=1}^{n} \left(\frac{j+1}{j}\right)^3 = (n+1)^3.$$

This proves (4). Moreover $n^A B_n \to 0$ for every fixed real A, since its logarithm is $A \log n + 3 \log(n+1) - (m+1) \log(3/2) \to -\infty$.

Proof of the asymptotic

Put $k = n - m$. For $0 \leq r \leq m$, monotonicity gives

$$\frac{h_{k,r}}{p_n} \leq \frac{h_{n-r,r}}{p_{n-r}} \leq \frac{h_{n,r}}{p_k}.$$

The omitted tail in the lower sum is at most T_n, because $k \leq n$. For the terms $r > m$ in (3), use $p_{n-r} \geq 2$ and $h_{n-r,r} \leq h_{n,r}$. Thus

$$\frac{H_k(1) - B_n}{p_n} \;\leq\; \widehat{W}_{\mathrm{per}}(n) \;\leq\; \frac{H_n(1)}{p_k} + \frac{B_n}{2}. \tag{5}$$

Now $k/n \to 1$. From (1),

$$\frac{nH_k(1)}{p_n} \sim e^{\gamma} \frac{\log p_k}{\log n} \longrightarrow e^{\gamma}, \qquad \frac{nH_n(1)}{p_k} \sim e^{\gamma} \frac{n \log p_n}{k \log k} \longrightarrow e^{\gamma}.$$

Both limits use $\log p_k \sim \log k \sim \log n$ and $\log p_n \sim \log n$. Finally $nB_n \to 0$, so multiplying (5) by n and applying the squeeze theorem proves Theorem 1. □

Scope and verification

The proof applies to the full perimeter ensemble with unrestricted positive parts. It uses positivity, an exact partition bijection, and classical prime asymptotics. It does not differentiate a cumulative error estimate. The companion verifier checks the indexing by exact partition enumeration, checks exact finite versions of the bounds, and records larger numerical examples. These checks support the derivation; the argument above proves the assertion for all sufficiently large n. Review was by the authoring agent only, without independent peer review or a proof assistant.

Proposed follow-up Tasks 2–5

Task 2: Give explicit constants C, n_0 such that, for all $n \geq n_0$,

$$\left| \widehat{W}_{\mathrm{per}}(n) - \frac{e^{\gamma}}{n} \right| \leq C \frac{\log\log n}{n \log n}.$$

Task 3: For each fixed $\beta > 1$, determine the leading asymptotic for $\sum_{\mathrm{per}(\lambda)=n} \widehat{N}(\lambda)^{-\beta}$, including its constant.

Task 4: Determine the leading constant for the reciprocal supernorm sum over partitions of perimeter n with distinct parts.

Task 5: Under the probability measure on perimeter-n partitions proportional to $\widehat{N}(\lambda)^{-1}$, determine the limiting distribution of the number of parts after appropriate centering and scaling.

References

[1] J. C. Lagarias and C. Sun, *Asymptotics of reciprocal supernorm partition statistics*, The Ramanujan Journal **65** (2024), 487–520, Journal version (DOI). Preprint: arXiv:2308.15303v1.